Collaborative and Federated Bayesian Optimization:
A Survey & Unifying Framework

IEEE BigData 2024 Paper: Collaborative and Federated Black-box Optimization: A Bayesian Optimization Perspective

Raed Al Kontar
Associate Professor, University of Michigan
Website: umich.edu/∼alkontar, Email: alkontar@umich.edu

Outline

1. Federated Analytics
2. Bayesian Optimization BO
3. Unifying Frameworks and what exists within
   - Part I. Global Decisions
   - Part II. Conditioned Local Decisions
   - Part III. Predictive Solution: Improving the Surrogate

Why federated learning (FL) ?

- Computational power of edge devices is steadily increasing

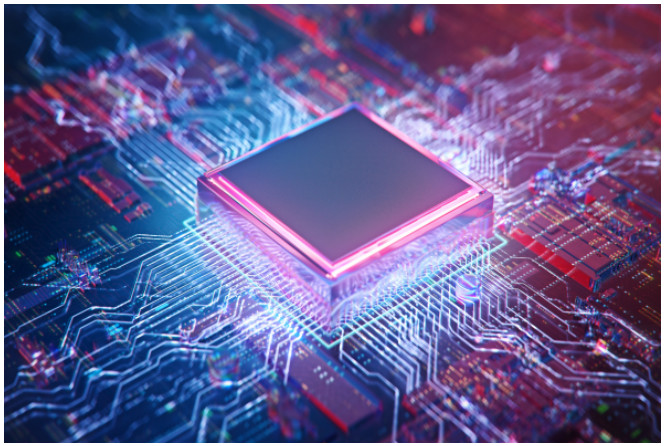
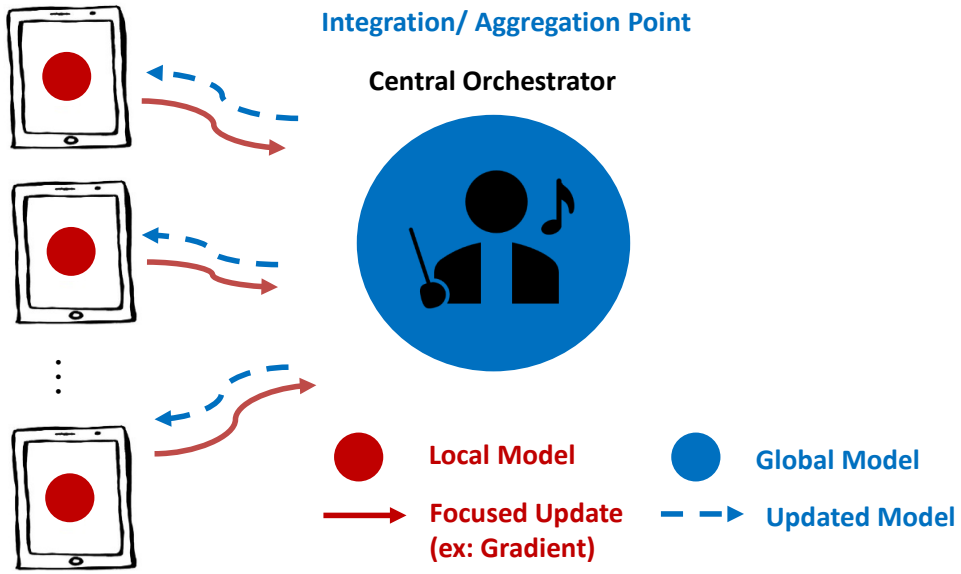


Figure 1: AI chips [6]

Federated Learning

Exploits edge compute resources to **process more of users' data where it's created** [9].

A lot has been done in FL



- Predictive: Mainly on deep learning
  - Faster convergence [8], improved aggregation [18], personalization [15], fairness [22], and uncertainty quantification [21, 17]. (ee [9] for a detailed review
- Descriptive: What is shared & unique ?
  - Federated low-rank decompositions such as personalized PCA [16], matrix completion [13, 14], and dictionary learning [10]

Bringing FL to prescription

Can we bring FL to a prescriptive front ?

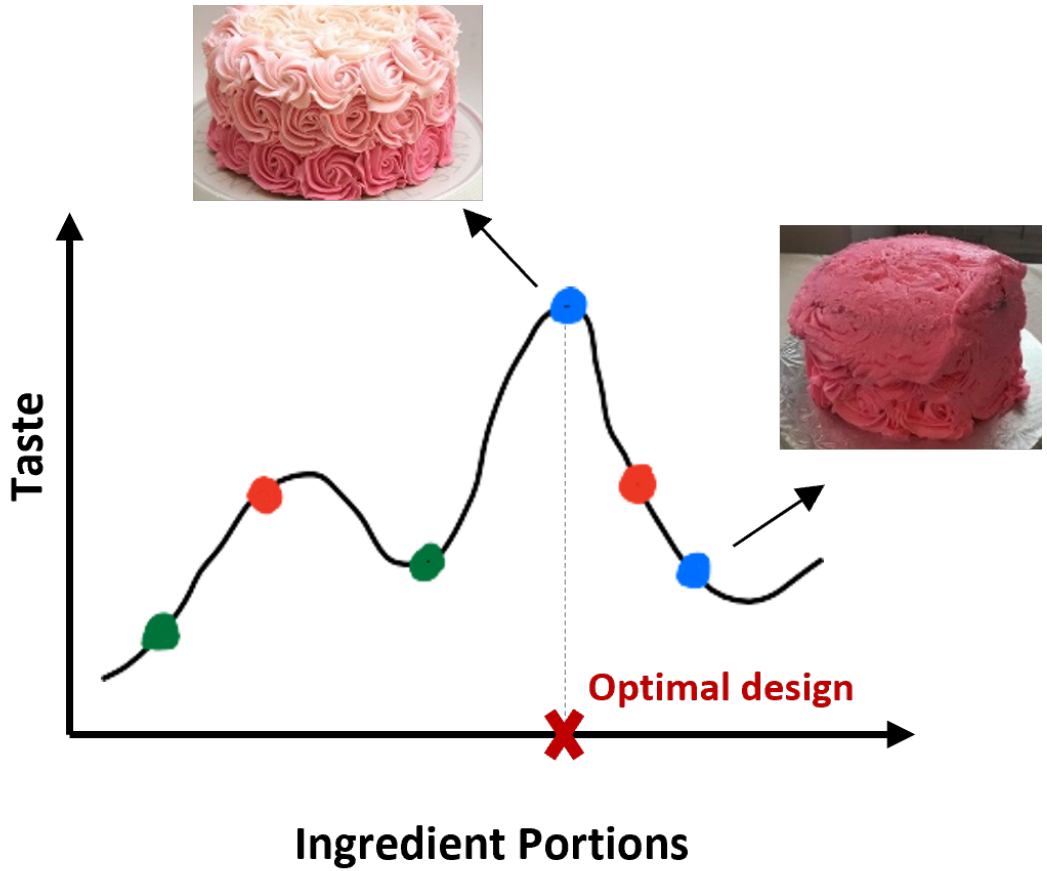
Goal

Provide a unifying framework and survey of prescriptive FL in the context of black-box optimization (BBOpt) —*an inherently sequential decision making problem*

- We focus on Bayesian optimization (BO) as tool for BBOpt
  - Current methods in this space are BO based

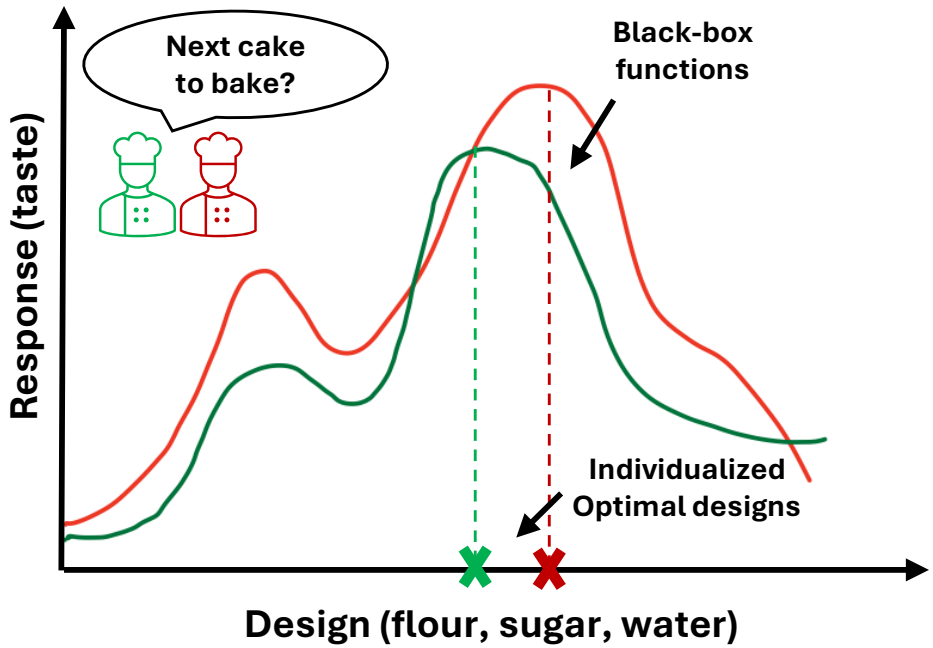
What is Bayesian optimization (BO) ?

- An approach for trial & error
- Decision: What experiment to run next ?



Challenge for collaborative & federated BO

- **How to collaborate ?**
- **How to collaborate in the presence of heterogeneity?**
  - Batch BO[1] does not work !



- **How to preserve the privacy of all collaborators?**

Setting notation: single client k

Assume a client k can run T experiments and starts with initial dataset $\mathcal{D}_{k,t} = \{(\boldsymbol{x}_{k,1}, y_{k,1}), \cdots, (\boldsymbol{x}_{k,t}, y_{k,t})\}$. At time t

- Decision: next-to-observe design $\boldsymbol{x}_{k,t}^{\text{new}} \in \mathbb{R}^d$
- Consequence: $y_k(\boldsymbol{x}) \triangleq f_k(\boldsymbol{x}) + \epsilon \in \mathbb{R}$
- Utility: $U(f_k(\boldsymbol{x}); \mathcal{D}_{k,t}) : \mathbb{R}^d \to \mathbb{R}$
- Surrogate: $\mathbb{P}_{\hat{f}_k|\mathcal{D}_{k,t}}$ for true function f_k

$$\boldsymbol{x}_{k,t}^{\text{new}} = \underset{\boldsymbol{x}}{\text{argmax}} \underbrace{F_{k,t}}_{\text{objective}} \triangleq \mathbb{E}_{\mathbb{P}_{\hat{f}_k|\mathcal{D}_{k,t}}} \left[U\left(f_k(\boldsymbol{x}); \mathcal{D}_{k,t}\right) \right]. \tag{1}$$

Part I. Global Decisions

Follow typical FL ?

K agents want to collaboratively optimize their black-box functions f_k

- Maximize average expected utility ($F_{k,t}$) across all agents

$$\max_{\boldsymbol{x}} \sum_{k=1}^{K} p_k F_{k,t}$$

  where p_k is some weight. Then solve with `FedAvg` [11] or others..
- **BAD IDEA** Global designs serve as initializations in predictive FL. All agents should not run same experiment

Goal

Collaboratively allow agents to decide on their individualized tests $\boldsymbol{x}_{k,t}^{\text{new}}$

Solution: Consensus [20]

- New formulation

$$\boldsymbol{x}_{k,t}^{\text{new}} = \left[(\boldsymbol{W}^{(t)} \otimes \boldsymbol{I}_d) \boldsymbol{x}_t^{\mathcal{C}} \right]_k \tag{2}$$

Solution: Consensus [20]

- New formulation

$$\boldsymbol{x}_{k,t}^{\text{new}} = \left[(\boldsymbol{W}^{(t)} \otimes \boldsymbol{I}_d) \boldsymbol{x}_t^{\mathcal{C}} \right]_k \tag{2}$$

$$\text{such that} \quad \boldsymbol{x}_{k,t}^{\text{c}} = \left[\boldsymbol{x}_t^{\mathcal{C}} \right]_k = \underset{\boldsymbol{x}_k}{\text{argmax}}\, F_{k,t} \,,$$

  where $[\cdot]_k$ denotes the kth block of candidate solutions $\boldsymbol{x}_{\mathcal{C}}^{(t)}$

- $\boldsymbol{W}^{(t)}$ is symmetric and **doubly-stochastic** consensus matrix of size $K \times K$ with non-negative elements

Interesting features

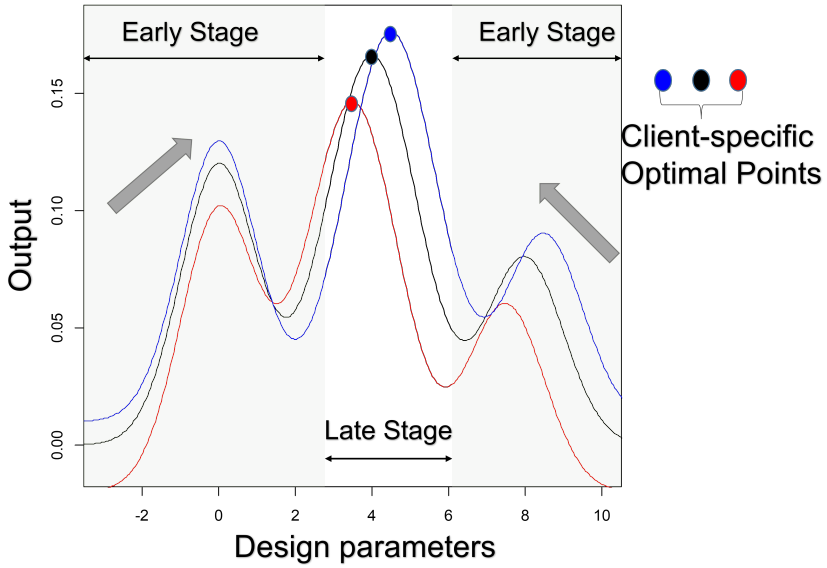
- $(\boldsymbol{W}^{(t)} \otimes \boldsymbol{I}_D)\boldsymbol{x}_{\mathcal{C}}$ naturally yields K solutions to all agents; one for each

Example when $d = 1$ and $K = 2$

$\boldsymbol{W}^{(t)} = \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix}$ and $\boldsymbol{x}^{\mathcal{C}} = (5, 7)$ then $\boldsymbol{W}^{(t)}\boldsymbol{x}^{\mathcal{C}} = (5.6, 6.4)^{\intercal}$. Client 1 and 2 run experiments at 5.6 and 6.4 respectively

- Naturally Distributed
- Only share input and not client specific output
- $\boldsymbol{W}^{(t)}$ can dynamically decide how much one client will affect the design choices for other clients
- Decision at time t in 1 communication round between cloud and agents

Designing the consensus matrix: one example



$$\boldsymbol{W}^{(t+1)} = \boldsymbol{W}^{(t)} + \begin{bmatrix} \frac{K-1}{TK} & -\frac{1}{TK} & \cdots & -\frac{1}{TK} \\ \vdots & \vdots & \vdots & \vdots \\ -\frac{1}{TK} & -\frac{1}{TK} & \cdots & \frac{K-1}{TK} \end{bmatrix},$$

Essential requirement

$\boldsymbol{W}^{(t)} \to \boldsymbol{I}$ for convergence under heterogeneity

Drawbacks & open questions

- Drawbacks
  - Active collaboration
  - Decisions dictated by central entity
  - Theory hard to derive
- Interesting questions
  - Privacy guarantees through sharing $\boldsymbol{x}^{\mathsf{c}}_{k,t} + \epsilon_t$
  - Different fidelities
  - Resource constraints

Part II. Conditioned Local Decisions

Central idea

- Agent k receives information $E_{k,t}$ from others agents $k' \in [K \setminus k]$. Then make decisions conditioned on this information

  - What should $E_{k,t}$ be ?
  - What should agent k do with $E_{k,t}$?

  The hope in FL is that $E_{k,t}$ is tiny in size but rich in information.

* will drop t for simpler notation

Conditioning the surrogate [2]

Agent k' shares $\boldsymbol{x}_k^+ \in E_k$ with Agent k if

$$\mu_k(\boldsymbol{x}) - \eta \cdot \sigma_{k'}(\boldsymbol{x}) > \delta_k \triangleq \max_{\boldsymbol{x}} \mu_k(\boldsymbol{x}) \,, \tag{3}$$

- LCB maximizer satisfies this constraint (if it exists)
- Comparison can be done privately (Yao's Millionaires' problem)

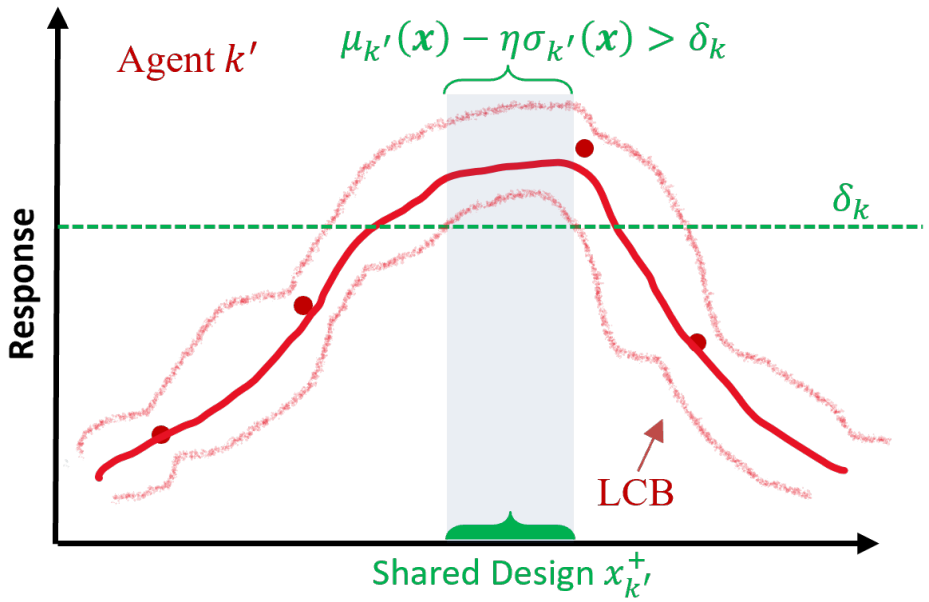


Figure 2: Agent k' shares a design from the shaded region with Agent k

Conditioning the surrogate [2]

- Agent k converts shared designs into constraints

$$\forall \boldsymbol{x}_{k'}^{+} \in E_k : f(\boldsymbol{x}_{k'}) > \delta_k$$

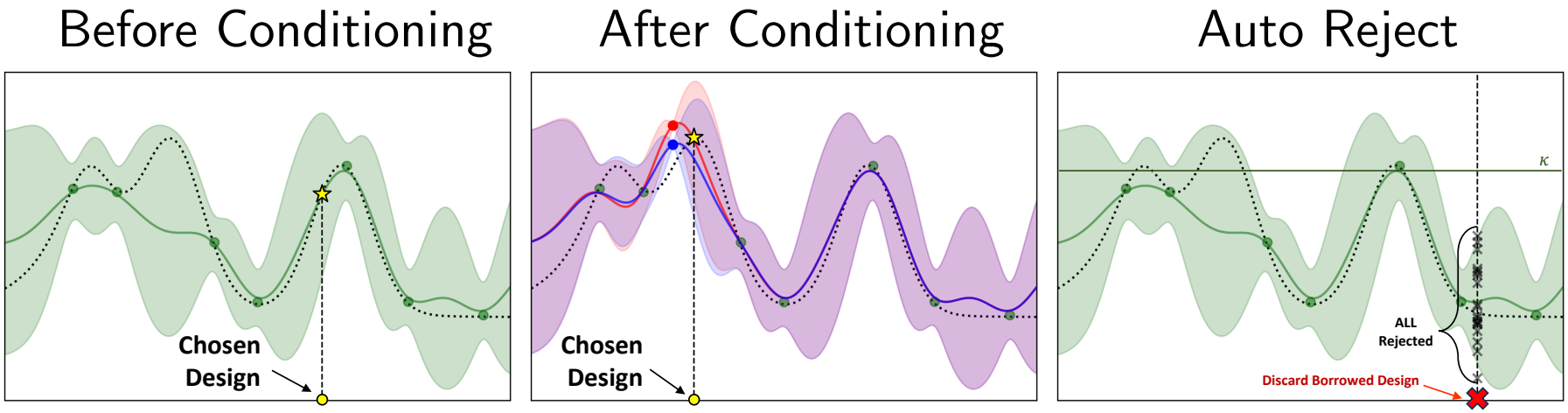
- Agent k learns the constrained $\mathcal{GP}$: $\mathcal{F}_k^{+} \triangleq \mathbb{P}_{\hat{f}_k|\mathcal{D}_k}|E_k$

$$F_k^{+} = p(\underbrace{f_k}_{\text{black-box}} | \underbrace{D_k}_{\text{own}}, \underbrace{E_k}_{\text{borrowed}})$$

- Agent k learns new utility $\mathbb{E}_{\mathcal{F}_k^{+}}[U(f(\boldsymbol{x}); \mathcal{D}_k)]$

Example via rejection sampling (RS)

- Sampling from Constrained $\mathcal{GP}$ and utility, can be done via RS
- **Guarantees adversarial convergence**



- **Challenge**: RS scales poorly

Conditioning the utility

- While not currently proposed we can let $E_k = \{\pi_{k'}\}_{k'}$, where $\pi_{k'}$ is a density shared by agent k'
  - Ex: $\pi_{k'} = \mathcal{N}(\boldsymbol{x}_{k'}^{+}, \sigma^2 \boldsymbol{I}_d)$
- Tweak utility similar to incorporating prior knowledge in πBO[7]

$$\boldsymbol{x}_{k,t}^{\text{new}} = \underset{\boldsymbol{x}}{\text{argmax}} \left[F_{k,t} \times \left(\prod_{k'} \pi_{k'} \right)^{\frac{\beta}{T}} \right]$$

- **Challenge**: What should β be ?

Sharing $\mathcal{GP}$ features [4, 5]

Using Random Fourier Features (RFFs) [12]:

$$\hat{f}_k(\boldsymbol{x}) \approx \boldsymbol{\phi}^\top(\boldsymbol{x})\boldsymbol{w}_k \quad \text{where} \quad \boldsymbol{w}_k \sim \mathbb{P}_{\boldsymbol{w}_k|\mathcal{D}_{k,t}}$$

- Agents $\{k'\}$ share $\boldsymbol{w}_{k'}^{(s)} \sim \mathbb{P}_{\boldsymbol{w}_{k'}|\mathcal{D}_{k',t}}$ with Agent k so $E_{k,t} = \{\boldsymbol{w}_{k'}^{(s)}\}$
- Local decisions for Agent k
  - Chose own $\boldsymbol{w}_k^{(s)}$ with probability and **increasing** p_t
  - Chose $\boldsymbol{w}_{k'}^{(s)}$ randomly sampled from $E_{k,t}$ with probability $1 - p_t$
  - Thompson Sampling (TS): $\boldsymbol{x}_{k,t}^{\text{new}} = \arg\max_{\boldsymbol{x}} \boldsymbol{\phi}^\top(\boldsymbol{x})\boldsymbol{w}^{\text{chosen}}$

Challenge: Only TS compatible

Part III. Predictive Solution: Improving the Surrogate

- Agents collaborate to get better surrogates
- Local decisions follow regular BO (1)
- Can potentially be incorporated within Part I and II

Federated $\mathcal{GP}$ [19]

- $f_k(\boldsymbol{x}) \sim \mathcal{GP}(0, \mathcal{K}(\cdot, \cdot; \boldsymbol{\theta}_{\mathcal{K}}))$
- $y_k = f_k(\boldsymbol{x}) + \epsilon_k$, $\epsilon \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$
- $\boldsymbol{\theta} = (\boldsymbol{\theta}_{\mathcal{K}}, \sigma^2)$

Federated learning of $\boldsymbol{\theta}$. Can we `FedAvg` or recent forms ?

$$\underset{\boldsymbol{\theta}}{\text{argmax}} \sum_{k=1}^{K} p_k \underbrace{L_{k,t}}_{\text{local objective}} \triangleq \sum_{k=1}^{K} p_k \log \mathbb{P}(\underbrace{\boldsymbol{y}_{k,[t]}}_{\text{local data at } t} | \mathcal{D}_{k,t}; \boldsymbol{\theta})$$

- **Unlike neural networks, $\mathcal{GP}$ gradients are biased !**
- [19] show that you can still converge (with fast decaying correlations)
- Equivalent to learning a global prior where prediction are obtained by conditioning on local data $\mathcal{D}_{k,t}$

Federated Multi-output $\mathcal{GP}$ [3]

Give flexibility by letting agents have shared $\boldsymbol{\theta}^g$ and unique parameters $\boldsymbol{\theta}^k$

$$[f_1(\boldsymbol{x}), \cdots, f_K(\boldsymbol{x})]^\top \sim \mathcal{MGP}(\mathbf{0}, \text{cov}^f_{k,k'}(\boldsymbol{x}, \boldsymbol{x}'))$$

- **Cannot be written as sum over agent objectives**
- Using Variational Inference, [3] show that $\mathcal{MGP}$ log-likelihood $\approx$

$$\sum_{k=1}^{K} p_k \tilde{L}_{k,t} \quad \text{where} \quad \tilde{L}_{k,t}\text{'s are independent} \tag{4}$$

- `FedAvg`-like approach to update $\{\{\boldsymbol{\theta}^k\}, \sigma^2, \boldsymbol{\theta}^g\}$
  - Only requires sharing $\boldsymbol{\theta}^g$

Open questions

- Initializations
- Learning Heterogeneity and whether to collaborate and when to stop?
  - More exploration

References

References I

[1] Azimi, J., Fern, A., and Fern, X. (2010). Batch bayesian optimization via simulation matching. *Advances in Neural Information Processing Systems*, 23.

[2] Chen, Q., Jiang, L., Qin, H., and Kontar, R. A. (2024). Multi-agent collaborative bayesian optimization via constrained gaussian processes. *Technometrics*, (just-accepted):1–23.

[3] Chung, S. and Al Kontar, R. (2024). Federated multi-output gaussian processes. *Technometrics*, 66(1):90–103.

[4] Dai, Z., Low, B. K. H., and Jaillet, P. (2020). Federated bayesian optimization via thompson sampling. *Advances in Neural Information Processing Systems*, 33:9687–9699.

[5] Dai, Z., Low, B. K. H., and Jaillet, P. (2021). Differentially private federated bayesian optimization with distributed exploration. *Advances in Neural Information Processing Systems*, 34:9125–9139.

References II

[6] Gray, P. and Pancewicz, S. (2020). The ai chips race is on – what role will ip play? `https://www.eenewseurope.com/news/ai-chips-race-what-role-will-ip-play`. Accessed: 2020-07-18.

[7] Hvarfner, C., Stoll, D., Souza, A., Lindauer, M., Hutter, F., and Nardi, L. (2022). pibo: Augmenting acquisition functions with user beliefs for bayesian optimization. *arXiv preprint arXiv:2204.11051*.

[8] Karimireddy, S. P., Kale, S., Mohri, M., Reddi, S., Stich, S., and Suresh, A. T. (2020). Scaffold: Stochastic controlled averaging for federated learning. In *ICML*, pages 5132–5143. PMLR.

[9] Kontar, R., Shi, N., Yue, X., Chung, S., Byon, E., Chowdhury, M., Jin, J., Kontar, W., Masoud, N., and Nouiehed, M. (2021). The internet of federated things (ioft). *IEEE Access*, 9:156071–156113.

[10] Liang, G., Shi, N., Al Kontar, R., and Fattahi, S. (2024). Personalized dictionary learning for heterogeneous datasets. *NeurIPS*, 36.

References III

[11] McMahan, B., Moore, E., Ramage, D., Hampson, S., and y Arcas, B. A. (2017). Communication-efficient learning of deep networks from decentralized data. In *Artificial Intelligence and Statistics*, pages 1273–1282.

[12] Rahimi, A. and Recht, B. (2007). Random features for large-scale kernel machines. *NeurIPS*, 20.

[13] Shi, N., Al Kontar, R., and Fattahi, S. (2023a). Heterogeneous matrix factorization: When features differ by datasets. *Submitted*.

[14] Shi, N., Fattahi, S., and Kontar, R. A. (2024). Triple component matrix factorization: Untangling global, local, and noisy components. *Journal of Machine Learning Research*.

[15] Shi, N. and Kontar, R. A. (2023). Personalized federated learning via domain adaptation with an application to distributed 3d printing. *Technometrics*, 65(3):328–339.

References IV

[16] Shi, N. and Kontar, R. A. (2024). Personalized pca: Decoupling shared and unique features. *Journal of Machine Learning Research*, 25(41):1–82.

[17] Shi, N., Lai, F., Al Kontar, R., and Chowdhury, M. (2023b). Fed-ensemble: Ensemble models in federated learning for improved generalization and uncertainty quantification. *IEEE Transactions on Automation Science and Engineering*.

[18] Wang, H., Yurochkin, M., Sun, Y., Papailiopoulos, D., and Khazaeni, Y. (2020). Federated learning with matched averaging. *ICLR*.

[19] Yue, X. and Kontar, R. (2024). Federated gaussian process: Convergence, automatic personalization and multi-fidelity modeling. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.

[20] Yue, X., Kontar, R. A., Berahas, A. S., Liu, Y., Zai, Z., Edgar, K., and Johnson, B. N. (2023a). Collaborative and distributed bayesian optimization via consensus: Showcasing the power of collaboration for optimal design. *arXiv preprint arXiv:2306.14348*.

References V

[21] Yue, X., Kontar, R. A., and Gómez, A. M. E. (2024). Federated data analytics: A study on linear models. *IISE Transactions*, 56(1):16–28.

[22] Yue, X., Nouiehed, M., and Al Kontar, R. (2023b). Gifair-fl: A framework for group and individual fairness in federated learning. *INFORMS Journal on Data Science*, 2(1):10–23.